



Hotel Booking Demand Analysis & Cancellation prediction

Data Science Use case

START

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Tools & Technologies

Tools & Technologies

1. Programming Languages –

- a. Preferably use Python or R
- b. If you are not comfortable with above, then can use any other programming languages

2. IDE –

- a. Preferably use Jupyter, Rstudio, Pycharm, Spyder, VS Code etc

3. Visualization/ EDA –

- a. Preferably use Python or R
- b. If you are not comfortable with above, then can use Power BI, Tableau, Excel or any other tools

4. Documentation

- a. Preferably use Markdown languages
- b. If you are not comfortable with above, then can use any other tools/languages

5. Final Report –

- a. Share your development code base (if in a separate script)
- b. Provide the final report and approach analysis in Jupyter notebook, Google Colab or R markdown
- c. If you are not comfortable with above, use your preferred reporting tool (Word, PDF etc)

Use Case – Hotel Booking

Hotel Booking - Overview

Have you ever wondered:

- When is the best time of the year to book a hotel room?
- What is the optimal length of stay in order to get the best daily rate?
- What if you wanted to predict whether or not a hotel is likely to receive a disproportionately high number of special requests?
- Whether a booking is going to be cancelled?

This hotel booking dataset can help you to explore those questions!



Hotel booking pain points from the perspectives of Hotel, Guest and Data Science

- Hotels currently don't know about the guests' preferences, booking / cancellation patterns, deeper insights etc.
- It would be nicer for the hotels to have a model to predict if a guest will come.



Hotel

- Descriptive analytics
Q1. Provide various descriptive analytics and insights
- Data Science Use Cases
Q2. Suggest the use cases suitable here from the perspective of the hotel owner?



Guest

- Data Science Use Cases
Q3. Suggest the use cases suitable here from the perspective of the guests?



Data Science

- Q4. Develop a model to predict bookings cancellation and explain the model as well as it's output.

Data

Hotel Dataset

Description

This data set contains booking information for a city hotel and a resort hotel and includes information such as when the booking was made, length of stay, the number of adults, children, and/or babies, and the number of available parking spaces, among other things.

Data description

Dataset Features

Dataset contains following features:

- 1.hotel
- 2.is_canceled
- 3.lead_time
- 4.arrival_date_year
- 5.arrival_date_month
- 6.arrival_date_week_number
- 7.arrival_date_day_of_month
- 8.stays_in_weekend_nights
- 9.stays_in_week_nights
- 10.adults
- 11.children
- 12.babies
- 13.meal
- 14.country
- 15.market_segment
- 16.distribution_channel
- 17.is_repeated_guest
- 18.previous_cancellations
- 19.previous_bookings_not_canceled
- 20.reserved_room_type
- 21.assigned_room_type
- 22.booking_changes
- 23.deposit_type
- 24.agent
- 25.company
- 26.days_in_waiting_list
- 27.customer_type
- 28.adr
- 29.required_car_parking_spaces
- 30.total_of_special_requests
- 31.reservation_status
- 32.reservation_status_date

Thank you for your passion!

